

Week 06 – Assignment

Scenario: Secret Notice Encoder

A college club regularly shares announcements with its members. Since these messages are posted on a public notice board, the club wants to make them difficult for outsiders to read. To achieve this, the club uses a Substitution Cipher. Every alphabetic character is replaced by the previous letter in `string.ascii_letters`. Non-alphabetic characters such as spaces, digits, and punctuation marks remain unchanged.

The following Python program performs this encryption.

```
1. import string
2.
3. cipher = {}
4.
5. letters = string.ascii_letters
6.
7. for i in range(len(letters)):
8.     cipher[letters[i]] = letters[i - 1]
9.
10. text = input("Enter a message: ")
11.
12. encrypted = ""
13.
14. for ch in text:
15.     if ch in cipher:
16.         encrypted += cipher[ch]
17.     else:
18.         encrypted += ch
19.
20. print("Original :", text)
21. print("Encrypted:", encrypted)
```

Q1. (MCQ)

Which line creates the substitution rule that maps each letter to its replacement?

- A. Line 5
- B. Line 7
- C. Line 8
- D. Line 16

Answer: C

Justification: Line 8 stores each original letter as a key and its substituted letter as the corresponding value in the dictionary.

Q2. (MCQ)

Suppose the user enters the input:

ABC123

Which statement ensures that the digits 1, 2, and 3 remain unchanged?

- A. Line 10
- B. Lines 17–18
- C. Line 20
- D. Line 21

Answer: B

Justification: If a character is not found in the dictionary (such as digits or punctuation), the else block copies it unchanged.

Q3. (MCQ)

What is the purpose of Line 12?

- A. It stores the substitution dictionary.
- B. It stores the original input.
- C. It initializes an empty string to build the encrypted message.
- D. It prints the encrypted output.

Answer: C

Justification: The variable encrypted starts as an empty string, and encrypted characters are appended to it during processing.

Q4. (MCQ)

If Line 16

encrypted += cipher[ch]

is replaced with

encrypted += ch

what will be the result?

- A. Only uppercase letters will be encrypted.
- B. The program will generate an error.

- C. Only digits will be encrypted.
- D. The output will be identical to the input.

Answer: D

Justification: The substitution dictionary is no longer used. Every character is copied directly into the output.

Q5. (MSQ)

Which of the following statements about the program are correct? (Select all that apply.)

- A. Line 5 stores all lowercase and uppercase English letters.
- B. Line 8 creates a substitution mapping using a dictionary.
- C. Line 15 checks whether the current character is a digit.
- D. Lines 20 and 21 display both the original and encrypted messages.

Answer: A, B, D

Justification:

- A: `string.ascii_letters` contains both lowercase and uppercase English letters.
- B: Line 8 creates the mapping from each letter to its substitute.
- C: Line 15 checks whether the character exists in the dictionary, not whether it is a digit.
- D: The final two print statements display both the original input and the encrypted output.

Scenario: Tic-Tac-Toe Winner Checker

A programming club is developing a simple Tic-Tac-Toe game. The game board is represented using a nested list, where each inner list represents a row of the board.

- 'X' represents Player X.
- 'O' represents Player O.
- ' ' (space) represents an empty cell.

The program checks only the rows of the board to determine whether either player has won.

If a row contains three identical non-space symbols, that player is declared the winner.

Otherwise, the program reports that there is no row winner.

```
1. board = [  
2.     ['X', 'X', 'X'],  
3.     ['O', ' ', 'O'],  
4.     [' ', 'O', ' ']  
5. ]  
6.  
7. winner = "None"  
8.  
9. for row in board:  
10.     if row[0] == row[1] == row[2] and row[0] != ' ':  
11.         winner = row[0]  
12.         break  
13.  
14. if winner != "None":  
15.     print("Winner:", winner)  
16. else:  
17.     print("No row winner")
```

Q1. (MCQ)

Which line checks whether all three cells in a row contain the same non-empty symbol?

- A. Line 7
- B. Line 9
- C. Line 10
- D. Line 15

Answer: C

Justification: Line 10 compares all three elements of a row and ensures that they are not blank.

Q2. (MCQ)

What is the purpose of the break statement in Line 12?

- A. It skips the current row.
- B. It exits the loop after finding a winner.
- C. It resets the board.
- D. It starts checking the rows again.

Answer: B

Justification: Once a winning row is found, there is no need to examine the remaining rows.

Q3. (MCQ)

Which variable stores the winning player's symbol?

- A. board
- B. row
- C. winner
- D. print

Answer: C

Justification: The variable winner is updated with 'X' or 'O' when a winning row is found.

Q4. (MCQ)

If Line 11

```
winner = row[0]
```

is replaced with

```
winner = "Player"
```

what will the program display when a winning row is found?

- A. The winning symbol (X or O)
- B. "Player"
- C. "None"
- D. The entire row

Answer: B

Justification: The variable winner would always contain the string "Player" instead of the winning symbol.

Q5. (MSQ)

Which of the following statements about the program are correct? (Select all that apply.)

- A. The board is represented using a nested list.
- B. The program checks only row-wise winning conditions.
- C. The loop stops immediately after a winner is found.
- D. The program checks both row-wise and diagonal winning conditions.

Answer: A, B, C

Justification:

- A: Correct. The board is stored as a list of lists.
- B: Correct. The program examines only the rows for a winning condition.
- C: Correct. The break statement terminates the loop once a winner is detected.

- D: Incorrect. The program does not check diagonal or column-wise winning conditions.

Scenario-Based Question: Recursive Folder Search

A company stores its documents in a hierarchy of folders. Each folder can contain:

- Files
- Other folders

A software engineer is asked to write a program to find a file named Report.pdf. The program begins with the main folder. Whenever it encounters a subfolder, it performs the same search process inside that subfolder. If the file is found, the search stops. If a folder contains no more subfolders or files, the program returns to the previous folder and continues searching.

No loops are explicitly used to revisit folders; instead, the same search procedure is repeatedly applied to each subfolder.

Questions

Q1. (MCQ)

Which characteristic of the search process best represents recursion?

- A. The search visits folders in alphabetical order.
- B. The same search procedure is applied repeatedly to each subfolder.
- C. Every folder contains the same number of files.
- D. The search uses multiple computers simultaneously.

Answer: B

Justification: In recursion, the same task is performed repeatedly on smaller instances of the same problem.

Q2. (MCQ)

In this scenario, what is the base case for the recursive search?

- A. The search begins from the main folder.
- B. A folder contains only PDF files.
- C. A folder has no more files or subfolders to search, or the required file is found.
- D. The search enters a subfolder.

Answer: C

Justification: A recursive process stops when it reaches a condition where no further recursive calls are needed.

Q3. (MCQ)

Why is recursion a suitable approach for this problem?

- A. Every folder has exactly one subfolder.
- B. The same search operation must be repeated for each subfolder.
- C. Recursion is always faster than iteration.
- D. Files can only exist in the main folder.

Answer: B

Justification: Each subfolder is simply a smaller version of the original problem, making recursion a natural solution.

Q4. (MCQ)

What would most likely happen if the search process had no base case?

- A. The search would immediately stop.
- B. The search would examine only the main folder.
- C. The search could continue indefinitely or until system resources are exhausted.
- D. The search would automatically switch to an iterative approach.

Answer: C

Justification: Without a stopping condition, recursive calls continue indefinitely, eventually causing a stack overflow or exhausting system resources.

Q5. (MSQ)

Which of the following statements correctly describe recursion in the given scenario? (Select all that apply.)

- A. The same search procedure is used for every subfolder.
- B. The search must have a stopping condition.
- C. Each subfolder represents a smaller instance of the original problem.
- D. Recursion can only be used if every folder contains exactly one subfolder.

Answer: A, B, C

Justification:

- A: Correct. The same search logic is applied to every subfolder.
- B: Correct. Every recursive solution requires a base case to terminate.
- C: Correct. Each subfolder is a smaller version of the original folder search problem.

- D: Incorrect. A folder may contain any number of subfolders or none at all; recursion does not require exactly one subfolder.

Scenario: Recursive Binary Search for Student Roll Numbers

A university stores student roll numbers in ascending order. When an administrator wants to verify whether a particular roll number exists, the system performs a recursive binary search. The search begins by checking the middle roll number in the current range.

- If the middle roll number matches the required roll number, the search is successful.
- If the required roll number is smaller, the same search process is repeated on the left half.
- If the required roll number is larger, the search continues recursively on the right half.
- If the search range becomes invalid (the left index becomes greater than the right index), the roll number is reported as not found.

The following program implements this recursive binary search.

```
1. def binary_search(arr, low, high, target):
2.     if low > high:
3.         return -1
4.
5.     mid = (low + high) // 2
6.
7.     if arr[mid] == target:
8.         return mid
9.
10.    elif target < arr[mid]:
11.        return binary_search(arr, low, mid - 1, target)
12.
13.    else:
14.        return binary_search(arr, mid + 1, high, target)
15.
16. numbers = [105, 112, 118, 124, 131, 145, 152, 168]
17. key = 131
18.
19. result = binary_search(numbers, 0, len(numbers)-1, key)
20.
21. if result != -1:
22.     print("Roll number found at index", result)
23. else:
24.     print("Roll number not found")
```

Q1. (MCQ)

Which line represents the base case where the search terminates because the search range has become invalid?

- A. Line 5
- B. Line 2
- C. Line 7
- D. Line 14

Answer: B

Justification: Line 2 checks whether $low > high$. If true, there are no more elements to search, and the recursion terminates.

Q2. (MCQ)

Which line performs the recursive search on the left half of the list?

- A. Line 5
- B. Line 8
- C. Line 11
- D. Line 19

Answer: C

Justification: When the target is smaller than the middle element, the function recursively searches from low to $mid - 1$.

Q3. (MCQ)

What is the purpose of Line 5?

- A. It checks whether the element is found.
- B. It calculates the middle index of the current search range.
- C. It sorts the list.
- D. It prints the search result.

Answer: B

Justification: The middle index is calculated so that the current search range can be divided into two halves.

Q4. (MCQ)

If Line 14

return binary_search(arr, mid + 1, high, target)

is replaced with

return binary_search(arr, low, high, target)

what is the most likely outcome when the target is greater than the middle element?

- A. The program always finds the target immediately.
- B. The search correctly moves to the right half.
- C. The search continues without reducing the search range, leading to infinite recursion.
- D. The program searches only the left half.

Answer: C

Justification: Since the search boundaries do not change, the same recursive call is repeated indefinitely.

Q5. (MSQ)

Which of the following statements about the program are correct? (Select all that apply.)

- A. The recursive calls in Lines 11 and 14 reduce the search space in every recursive step.
- B. The program can correctly perform binary search even if numbers is not sorted.
- C. Line 5 compares the target with every element in the list.
- D. The recursion stops either when the target is found (Line 8) or when the search range becomes invalid (Line 3).

Answer: A, D

Justification:

- A: Correct. Each recursive call searches only one half of the remaining list.
- B: Incorrect. Binary search requires the list to be sorted.
- C: Incorrect. Line 5 only computes the middle index; it does not compare every element.
- D: Correct. These are the two terminating conditions of the recursive binary search.